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▼ (1) Course Introduction

[Module 1 Overview](#)[What is Machine Learning?](#)[Examples, Models, and Accuracy.](#)[Machine Learning Terminology.](#)

**Measure of Model Performance - Introduction****Measure of Model Performance - Linear Algebra review****Measure of Model Performance - Supervised Model Introduction****Measure of Model Performance - Create Operating Characteristic Curves****Measure of Model Performance - kNN****Module 1 Quiz**

May 30 100 pts

▼ (2) Linear Perceptron Algorithm**Module 2 Overview****Example & Representation****Learning Algorithm****Activation Functions****Multilayer Perceptrons****Linear Perceptron Example****Module 2 Quiz**



Jun 3 100 pts

▼ (3) Linear Regression**Module 3 Overview****Linear regression - Example****Linear regression - Representation****Linear regression - Evaluation****Optimization criterion: Deriving RSS error from scalar to matrix/vector notation****Closed-form solution: Derivation****Closed-form solution: Proof of convexity****Gradient descent: General formulation****Gradient descent for linear regression (batch, mini-batch, stochastic)****Non-linear regression: Representation, optimization****Non-linear Regression: Preliminary Discussion on Regularization****Module 3 Quiz**

Jun 6 100 pts

▼ (4) Logistic Regression, Regularization, & Feature Selection



Module 4 Overview



Introduction to Logistic Regression



Logistic Regression, Video 2



Logistic Regression, Video 3



Logistic Regression, Video 4



Logistic Regression, Video 5



Cross Validation



Cross-validation - Motivation



Cross-validation - Approaches



Cross Validation, Logistic Regression, and Python



Regularization and Ridge Regression, Video 2



Ridge Regression, Video 3

Lasso Regularization, Video 4

**LASSO Regularization, Video 5****LASSO Regularization, Video 6****LASSO Regularization, Video 7****Bootstrapping Ridge Regression and Lasso, Video 2****Bootstrapping Ridge and LASSO regression, video 1****Module 4 Quiz**

Jun 10 100 pts

**Homework 1: Linear Models (Code)**

Jun 10 54 pts

**Homework 1: Linear Models (Report)**

Jun 10 46 pts

▼ (5) Decision Trees, Random Forests, & Boosting**Module 5 Overview****Introduction to Decision Trees and Random Forests****Stumps and Partitioning****Decision Trees and Pruning**

**Classification and Regression Trees (CART)****Bagging and Random Forests****Out of Bag Error****Random Forest****Boosting Motivation****Boosting****Gradient Boosting****AdaBoost****Gradient Descent Boosting****Coding with XGBoost****Optional: Decision Tree****Optional: Regression Tree****Optional: Classification Tree****Optional: Bagging**

**Optional: Random Forest****Module 5 Quiz**

Jun 24 100 pts

**Homework 2: Regularization, Trees and Boosting (Code)**

Jun 24 70 pts

**Homework 2: Regularization, Trees and Boosting (Report)**

Jun 24 30 pts

▼ (6) SVMs, Kernels, & Hilbert Spaces**Module 6 Overview****Lagrange Optimization****Linearly Separable Case****Non-separable Case****Hinge Loss****Multi-Class SVMs****What have we learnt so far****Module 6 Quiz**

Jul 1 100 pts

Homework 3: SVM (Code)



Jul 1 50 pts

**Homework 3: SVM (Report)**

Jul 1 50 pts

▼ (7) Neural Networks & Deep Learning**Module 7 Overview****Perceptron & Connection with Logistic Regression****Representation & cost function of multilayer perceptron****Backpropagation Outline & Example****Practical issues with multilayer perceptron****Activation Function****Backpropagation****Optimization****Neural Network Training and Design****History of Deep Learning****Convolutional Neural Networks**

**Recurrent Neural Networks****Transfer Learning****Multi-task Learning****Deep Generative Models****Explainable AI - Interpretability****Module 7 Quiz**

Jul 15 100 pts

**Homework 4: Convolutional Neural Networks (Code)**

Jul 15 80 pts

**Homework 4: Convolutional Neural Networks (Report)**

Jul 15 20 pts

▼ (8) Dimensionality Reduction via Transformation (PCA, LDA, Autoencoders)**Module 8 Overview****Motivation for Dimensionality Reduction****Feature Selection: Wrappers****Feature Selection: Filters and Embedded Methods****Feature Transformation: Principal Component Analysis, Part 1**



Feature Transformation: Principal Component Analysis, Part 2



Proof Outline of Principal Component Analysis



Feature Transformation: Autoencoders and Clustering



Module 8 Quiz

Jul 22 100 pts

▼ (9) Clustering and Unsupervised Learning



Module 9 Overview



Overview & Basics



K-means Clustering



Gaussian Mixture Models - Description



Gaussian Mixture Models - Example



Hierarchical Clustering - Description



Hierarchical Clustering - Example



Module 9 Quiz

Jul 29 100 pts

▼ (10) Reinforcement Learning



Module 10 Overview



What is Reinforcement Learning?



An Example and Markov Decision Process



Value Iteration



Q Learning and Policy Iteration



Module 10 Quiz

Aug 5 100 pts



Homework 5: LSTM and Transformers

Aug 5 100 pts



Homework 5: Report

Aug 5 25 pts